Title (Units): **COMP7200 Blockchain and Cryptocurrencies (3,2,1)**

Course Aims: To introduce the concepts and applications of the blockchain technology; To introduce concepts and applications of two cryptocurrencies, Bitcoin and Ethereum; To explain the latest techniques of cryptocurrency and permissionless as well as permissioned blockchains; To conduct case studies to apply the blockchain technology to real-world applications.

Prerequisite: Nil

Course Intended Learning Outcomes (CILOs):
Upon successful completion of this course, students should be able to:

No.	Course Intended Learning Outcomes (CILOs)
	Knowledge
1	Compare different types of blockchains and cryptocurrencies
2	Identify the concepts and techniques of the cryptocurrency and consensus protocols
3	Describe the concepts and techniques of the smart contract and hyperledger
4	Investigate emerging techniques of the advanced topics on blockchain and cryptocurrencies
	Professional Skill
5	Design appropriate solutions to blockchain applications
6	Develop real-world applications using the blockchain technology

Calendar Description: This course provides an in-depth understanding and hands-on experience of the blockchain technology and cryptocurrencies. Students will learn the concepts and techniques behind different types of blockchains and cryptocurrencies, including mining mechanics, consensus protocols, smart contracts, and so on. Students will also learn blockchain application development skills via guided labs and projects.

Teaching and Learning Activities (TLAs):

CILOs	Type of TLA
1-4	Students will acquire the knowledge and skills on the blockchain technology and cryptocurrencies through lectures.
2-5	Students will learn the development skills of blockchain applications especially in fungible tokens and non-fungible tokens (NFT) via tutorials, assignments, quizzes, and guided labs.
5-6	Students will work on a group project to gain application development experience.

Assessment:

No.	Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
1	Continuous Assessment	40%	1-6	Assignments, quizzes and laboratories are designed to evaluate students' mastery of cryptocurrency concepts and techniques. A group project is used to assess their problem solving and application development skills.
2	Examination	60%	1-5	Final examination questions are designed to assess how well students understand and utilize the knowledge acquired.

Assessment Rubrics:

	Excellent (A)	Good (B)	Satisfactory (C)	Fail (F)
Describe different types of blockchains and their cryptocurrency applications	Thorough description of almost all types and applications	Description of most of the types and applications	Description of some of the types and applications	Description of a limited number of types and applications

Explain the concepts and techniques of the permissionless blockchain and cryptocurrency	Thorough explanation of almost all concepts and techniques	Explanation of most of the concepts and techniques	Explanation of some of the concepts and techniques	Explanation of a limited number of concepts and techniques
Explain the concepts and techniques of the permissioned blockchain	Thorough explanation of almost all concepts and techniques	Explanation of most of the concepts and techniques	Explanation of some of the concepts and techniques	Explanation of a limited number of concepts and techniques
Suggest appropriate solutions to blockchain applications	Suggestion of almost all correct solutions	Suggestion of most of the solutions	Suggestion of some of the solutions	Suggestion of very small number of solutions
Develop real-world applications using the blockchain technology	Thorough development of an application	Development of most parts of the application	Development of some parts of the application	Development of a very small part of the application

Course Content and CILOs Mapping:

Content		**CILO No.**
I	Introduction to Blockchains and Cryptography	1
II	Cryptocurrency	2,5,6
III	Cryptocurrency on Permissionless Blockchains	2,5,6
IV	Smart Contract and Decentralized Applications	3,5,6
V	Permissioned Blockchain	3,5,6
VI	Advanced Topics on Blockchain and Cryptocurrencies	4,5

References:

- Tobias Fertig and Andreas Schütz, "Blockchain: The Comprehensive Guide to Blockchain Development, Ethereum, Solidity, and Smart Contracts", Rheinwerk Computing, 2024
- Marco Di Maggio, "Blockchain, Crypto and DeFi: Bridging Finance and Technology" (1st ed), Wiley, 2024
- Mary C. Lacity and Steven C. Lupien, "Blockchain Fundamentals for Web 3.0", Epic Books, 2022.
- Imran Bashir. Mastering Blockchain – Third Edition, Packt Publishing, 2020.
- Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder. Bitcoin and Cryptocurrency Technologies, Princeton University Press, 2016.
- Arshdeep Bahga and Vijay Madisetti. Blockchain Applications: A Hands-On Approach, VPT, 2017.
- Alan T. Norman. Blockchain Technology Explained: The Ultimate Beginner's Guide About Blockchain Wallet, Mining, Bitcoin, Ethereum, Litecoin, Zcash, Monero, Ripple, Dash, IOTA And Smart Contracts, CreateSpace Independent Publishing Platform, 2017.
- Henning Diedrich. Ethereum: Blockchains, Digital Assets, Smart Contracts, Decentralized Autonomous Organizations. CreateSpace Independent Publishing Platform, 2016.
- Andreas M. Antonopoulos and Gavin Wood, Mastering Ethereum: Building Smart Contracts and DApps, O'Reilly Media; 2018.

Course Content:

<u>Topic</u>

I. Introduction to Blockchains and Cryptography
   A. Overview of Blockchain
   B. Types of Blockchains
   C. Blockchain Applications
   D. Cryptographic Primitives
   E. Distributed Systems

F. Consensus Protocols

II. Cryptocurrency
- A. Overview of Cryptocurrency
- B. Bitcoin basics
- C. Bitcoin implementation details
- D. Bitcoin's variants

III. Cryptocurrency on Permissionless Blockchains
- A. Ethereum basics
- B. Ethereum implementation details

IV. Smart Contract and Decentralized Applications
- A. Ethereum Virtual Machine
- B. Smart Contract, Dapp, DAO, Token, etc
- C. Developing Decentralized Applications in Ethereum

V. Permissioned Blockchain
- A. Hyperledger Fabric
- B. Applications of Hyperledger Fabric

VI. Advanced Topics on Blockchain and Cryptocurrencies
- A. Decentralized Finance (DeFi)
- B. Anonymity
- C. Privacy Protection
- D. Scalability: Lightning Network, Sharding, etc